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ASD ABSORB BOUNDARY CONDITION

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ASD absorb Boundary condition

by BuuNguyen » Wed Aug 13, 2025 5:01 am

Hi STKO team,

I noticed that you were updating the ASD Absorbing boundary condition. I am curious that:

For PDMY material,

1/ G values change during running, depending on the strain level (nonlinearity effect). Have the new ASD absorbed BC included the variation of G values during running?

2/ G also varies with depth. Should we use the average value of G for absorbing material, or have to divide the layer into sublayers for more accuracy?

3/ Additionally, I found this link, which mentions the theory of the ASD Absorbed BC ([https://opensees.github.io/OpenSeesDocu ...ndary.html](https://opensees.github.io/OpenSeesDocu...ndary.html))

Is there any link that explains the changed or updated information of this material?

Thank you for your support,
Buu



BuuNguyen



Posts: 134

Joined: Fri Oct 29, 2021 3:28 am

Location: Paraguay

Contact:



Re: ASD absorb Boundary condition

by STKO Team » Mon Aug 25, 2025 8:55 am

The absorbing boundary condition is based on linear elasticity theory.

However, we are now trying to update the G values in the absorbing



STKO Team



Posts: 3056

Joined: Tue Oct 29, 2019 8:45 am

boundary, using a Nonlinear material. To do so, in the new version you can assign a material rather than G, v, ρ values.

Note that it has been implemented (and currently beeing tested) on the 3D version only



Re: ASD absorb Boundary condition



by **Alessandro2016** » Mon Jun 29, 2026 8:37 am

Alessandro2016



Posts: 11

Joined: Tue Aug 19, 2025 1:44 pm

STKO Team wrote: ↑

Mon Aug 25, 2025 8:55 am

The absorbing boundary condition is based on linear elasticity theory.

However, we are now trying to update the G values in the absorbing boundary, using a Nonlinear material. To do so, in the new version you can assign a material rather than G, v, ρ values.

Note that it has been implemented (and currently beeing tested) on the 3D version only

Dear STKO team,

I am currently studying the `ASDAbsorbingBoundary3D` and `ASDAbsorbingBoundary3DAuto` elements in STKO/OpenSees, especially the possibility of accounting for nonlinear soil behavior in the absorbing boundary formulation.

In STKO, I noticed that there is a material option in `ASDAbsorbingBoundary3DMaterial`, where a previously defined 3D `nDMaterial` can be selected. My understanding is that the standard `ASDAbsorbingBoundary3D` element uses fixed parameters G , v , and ρ to define a linear-elastic free-field medium. However, I am not fully sure how the material option is currently implemented and supported.

When I tried to use this option with the solver from OpenSees-Solvers-3.4.0-Installer.exe, OpenSees returned the following error:

```
-----
ASDAbsorbingBoundary3D ERROR: Invalid optional flag "-mat".
Want: element ASDAbsorbingBoundary3D $tag $n1 $n2 $n3 $n4 $n5
      $n6 $n7 $n8 $G $v $rho $btype <-fx $tsxTag> <-fy $tsyTag> <-fz
      $tszTag>
-----
```

Therefore, I would like to ask for your clarification on the following points:

1. Is the material option in ASDAbsorbingBoundary3DMaterial currently supported in any officially released OpenSees/STKO solver version? If this feature has already been implemented, which OpenSees solver version should be used? For example, is it available only in OpenSees 3.7.0 or later?
2. When the material option is used, does it make the free-field columns fully nonlinear, or does it only use the selected nonlinear material to extract the current shear modulus G , which is then used to update the absorbing boundary parameters and dashpot coefficients?
3. Is this feature still under beta testing or active development?
4. What are the differences in modeling and application between manual ASDAbsorbingBoundary3D and the Auto version? Is the same nonlinear G -update feature available for the 2D element?
5. What is the correct approach for generating the initial stress field (gravity analysis) in the soil domain when using the manual ASDAbsorbingBoundary3D? Does it follow the same steps as a conventional model?

I await your reply at your earliest convenience, thank you very much for your help.

Best regards,
Alessandro



Re: ASD absorb Boundary condition

by **STKO Team** » Tue Jul 21, 2026 7:49 am

in that version it was not available yet.
download the 3.7.2



STKO Team



Posts: 3056
Joined: Tue Oct 29, 2019 8:45 am



Re: ASD absorb Boundary condition

by **Alessandro2016** » Tue Jul 21, 2026 3:42 pm

Thank you for your reply. I will use version 3.7.2 for the calculations. I would like to ask you about the "update the G values in the absorbing boundary" feature, is ASDAbsorbingBoundary2D element now available for use?
Best regards!



Alessandro2016



Posts: 11
Joined: Tue Aug 19, 2025 1:44 pm



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